

AAEC 4484/5484: Applied Economic Forecasting

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2/17/2026 Homework

This assignment aims to enhance your understanding of time series and data patterns. It is intended to be straightforward.

Instructions:

Please ensure that your graphs and visuals have proper titles and axis labels. Recall that you can use `help()` or `?seriesname` in your consoles to get general information on the dataset.

Question 1: Basic Visualization of Time Series Data

- i. Create time plots of the following time series: Bricks from `aus_production`, Lynx from `pelt`, Open for FB from `gafa_stock`, Demand for 2013 from `vic_elec`.

Note: Use the `wrap_plots()` function to arrange your plots in a 2x2 grid.

```
Plot1 <- aus_production %>% autoplot(Bricks) +
labs(title = "Australian Brick Quarterly Production",
y = "Millions of Bricks")

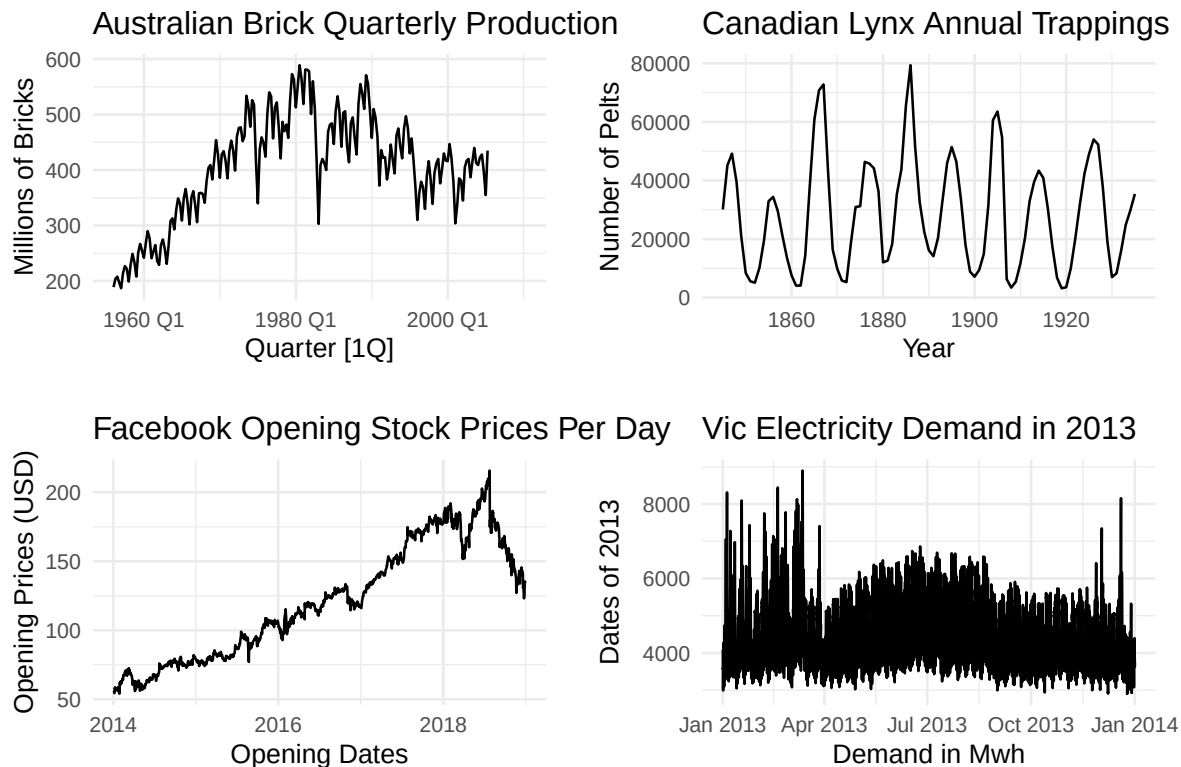
Plot2 <- pelt %>% autoplot(Lynx) +
labs(title = "Canadian Lynx Annual Trappings",
x = "Year",
y = "Number of Pelts")

Plot3 <- gafa_stock %>%
filter(Symbol == "FB") %>%
autoplot(Open) +
labs(title = "Facebook Opening Stock Prices Per Day",
y = "Opening Prices (USD)",
x = "Opening Dates" )

Plot4 <- vic_elec %>%
filter(year(Time) == 2013) %>%
autoplot(Demand) +
labs(title = "Vic Electricity Demand in 2013",
x = "Demand in Mwh",
y = "Dates of 2013")

graphlist <- list(Plot1,Plot2,Plot3,Plot4)

wrap_plots(graphlist, ncol = 2) &
theme_minimal(base_size = 10) +
theme(plot.margin = margin(10, 10, 10, 10, "pt"))
```



ii.

Briefly discuss any discernible pattern(s) you noticed in the data.

****Answer:** In the gafa_stock graph there seems to be upward trend or drift going on. That is until the middle of 2018 where it swoops back down. The lynx plot seems to have very large tops and bottoms in the graph with the first values being pretty close to the last observable values. The electricity demand plot shows a seasonal spread as you can see that the demand is higher during periods of more intense heat or cold. The brick plot has a seemingly recurring pattern in production in the what seems to be 5 to 7 year time span and could be evidence of a cycle going on.

Question 2: Random Data Sampling

In class, we explored the idea of drawing a random sample of data. Another practical operation is to randomly select a representative sample from a large dataset.

The `aus_retail` data set contains monthly retail turnover data for various industries and states in Australia. Your task is to randomly select an industry from this dataset and explore its time series patterns.

- i. In the code chunk below, set a seed of your choice to ensure a reproducible draw of the data. Next, use some combination of the `filter()` and `sample()` functions to draw a random sample of the data based on a single Series ID.

```
# Remove the `eval=FALSE` argument from the code chunk in order to run your codes.
set.seed(56) # Set this to your preferred seed

random_id <- aus_retail %>%
  distinct(`Series ID`) %>%
  slice_sample(n = 1) %>%
  pull(`Series ID`)

retail_sample <- aus_retail %>%
  filter(`Series ID` == random_id)
```

Use the `autoplot()`, `gg_season()`, `gg_subseries()`, and `ACF()` |> `autoplot()` functions to explore possible seasonality in your chosen sample. **Note: You can use the `theme(strip.text.y = element_blank())` to remove the y-axis labels in the `gg_subseries()` plot to make it look cleaner.**

- i.
 - Again use the `wrap_plot()` function with the `design` argument to arrange your plots as grids. **You are free to organize them however you wish.**

```
p5 <- retail_sample %>%
  autoplot(Turnover) +
  labs(title = "Time Plot of Retail Turnover", y = "Turnover")

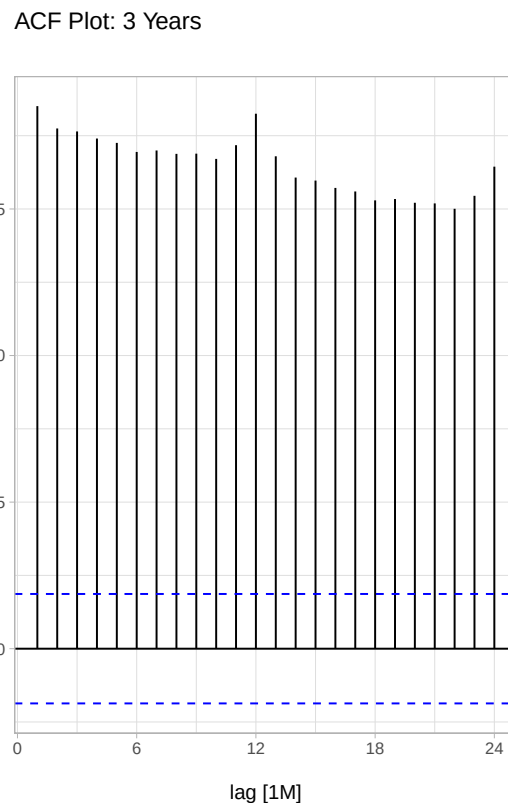
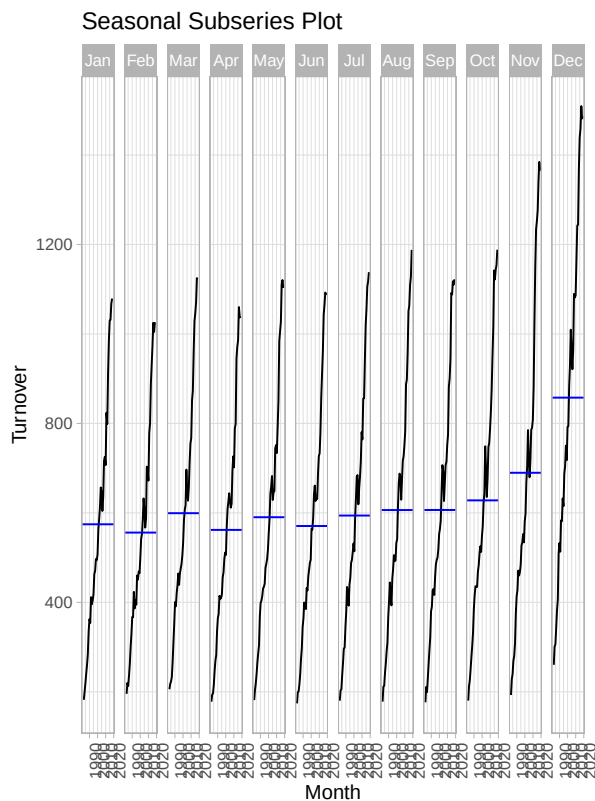
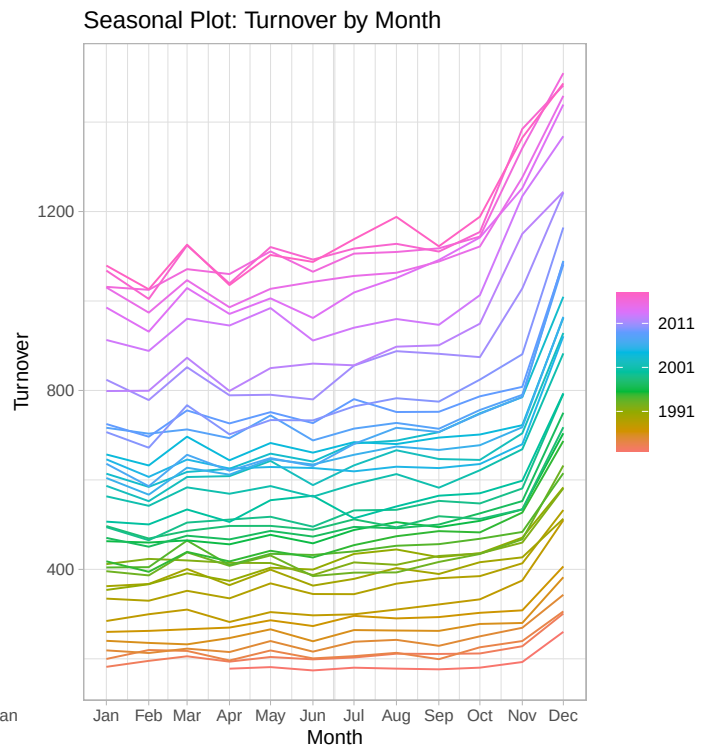
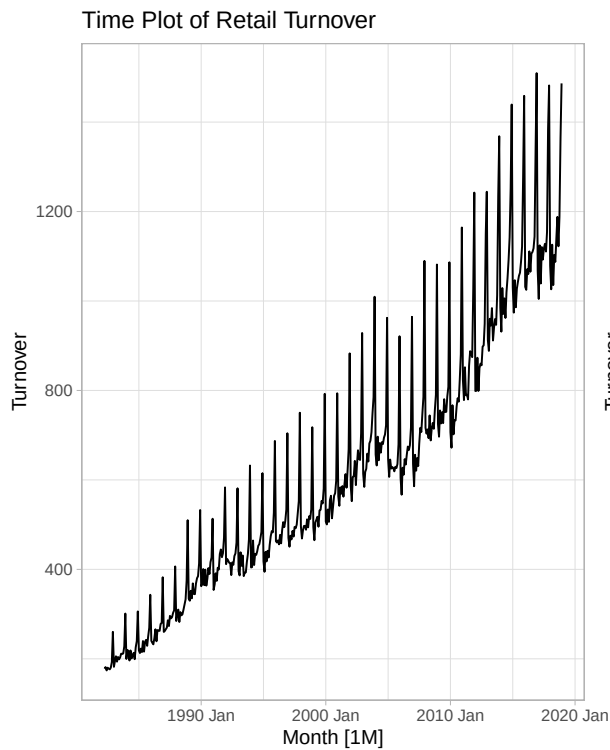
p6 <- retail_sample %>%
  gg_season(Turnover) +
  labs(title = "Seasonal Plot: Turnover by Month")

p7 <- retail_sample %>%
  gg_subseries(Turnover) +
  labs(title = "Seasonal Subseries Plot", y = "Turnover") +
  theme(strip.text.y = element_blank())

p8 <- retail_sample %>%
  ACF(Turnover, lag_max = 24) %>%
  autoplot() +
  labs(title = "ACF Plot: 3 Years")

randomlist <- list(p5, p6, p7, p8)

wrap_plots(randomlist,
  design = "AB
CD")
```



ii. What can you say about the series? Are there any seasonal patterns? Trends?

****Answer:** The Autocorrelation function seems to drift downward in the 3 year span of data with a relative maximum at 1 year. The autoplot as well does have a drift upwards over the years with a seemingly reliable pattern. The seasonal plot and seasonal subseries plots seem to have higher output in the winter months with march being larger than most as well. This could be sign of a high supply and demand in the beginning of the year and is also shown in the seasonal plot.

Question 3 - White Noise

The `aus_livestock` series contains data on monthly Meat production in Australia for human consumption.

- i. Using the `filter()` function, extract the number of **Lambs** slaughtered in **Victoria**. Store this variable as `lambs`.

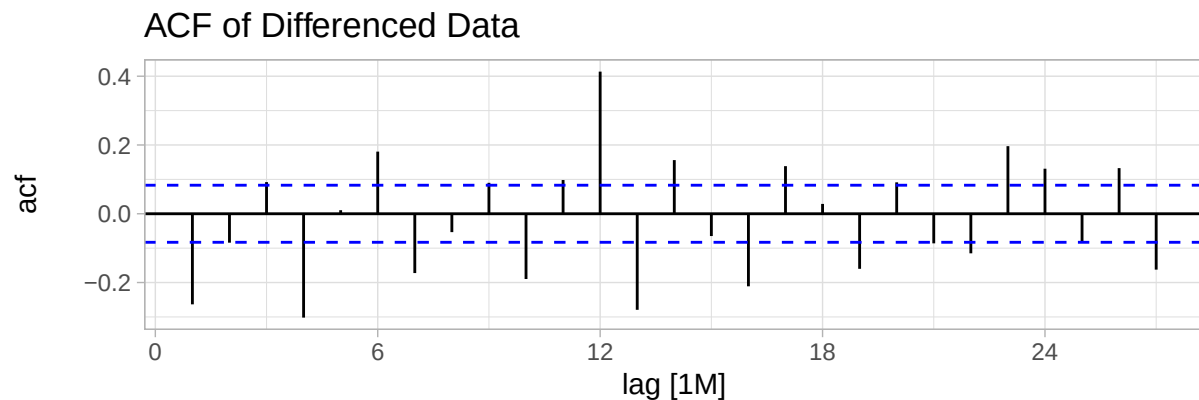
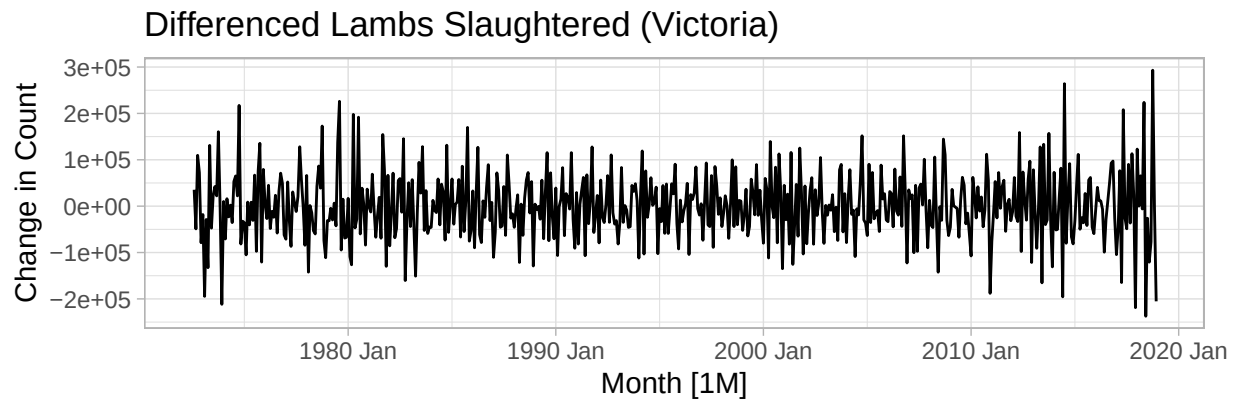
```
lambs <- aus_livestock %>% filter(Animal == "Lambs", State == "Victoria")
```

- ii. Now, using the `difference()` and `mutate()` functions, create a new column, `diff`, that computes the **month-to-month** changes (`lag = 1`) in your `lambs` series. Store this new data as `d.lambs`.

```
d.lambs <- lambs %>% mutate(diff = difference(Count, lag = 1))
```

- iii. Produce an `autoplot()` of `diff` along with the associated correlogram. Does the first differenced data now look like white noise? Did differencing remove any potential seasonality in the data? How about any trend? **Recall that a simple yes will not suffice. You will need to explain your conclusion.**

```
lambs <- aus_livestock %>%  
  filter(Animal == "Lambs", State == "Victoria")  
d.lambs <- lambs %>%  
  mutate(diff = difference(Count, lag = 1))  
c1 <- d.lambs %>%  
  autoplot(diff) +  
  labs(title = "Differenced Lambs Slaughtered (Victoria)",  
        y = "Change in Count")  
c2 <- d.lambs %>%  
  ACF(diff) %>%  
  autoplot() +  
  labs(title = "ACF of Differenced Data")  
c1/c2
```



**An-

swer: It does not look like white noise as there are many points where the lines go out of the confidence interval which lead me to believe there is more information to get out. There still seems to be a pattern of seasonality and there is a pattern of a line breaking the interval every 2-3 months but gradually lowers over time.

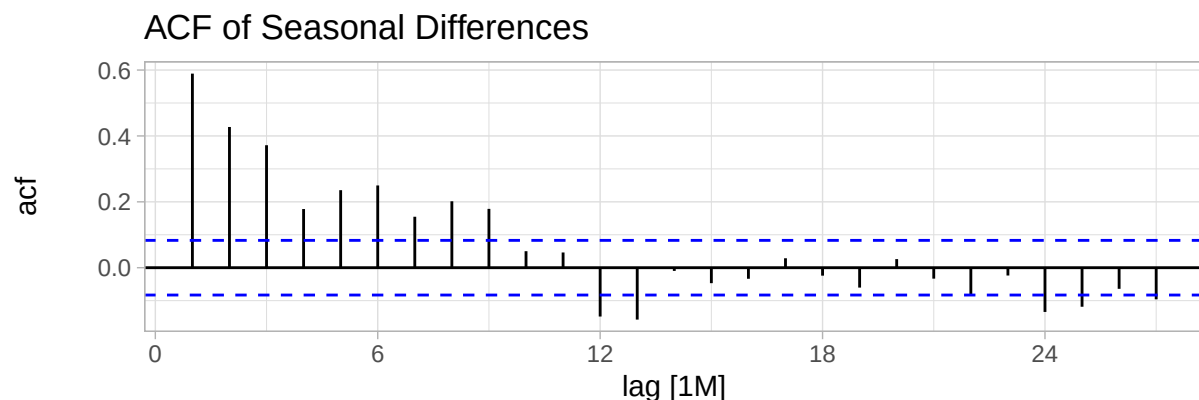
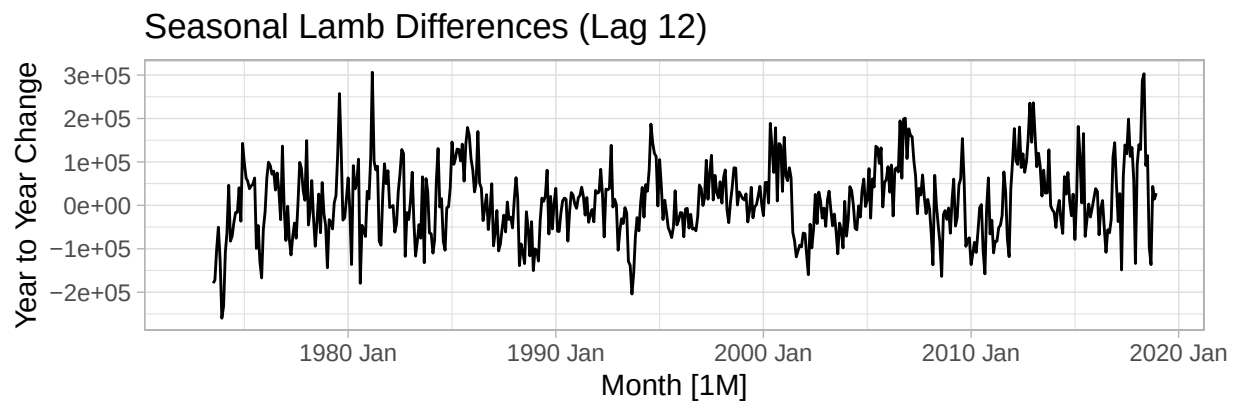
- iv. Return to the `lambs` series and again use the `difference()` and `mutate()` functions to create a new column called `diff12` that computes the **year-on-year** changes (`lag = 12`) in your `lambs` series instead. Store this new data as `d12.lambs`.

```
d12.lambs <- lambs %>% mutate(diff12 = difference(Count, lag = 12))
```

- v. Produce an autoplot of `diff12` along with the associated correlogram.

```
d12.lambs <- lambs %>%
  mutate(diff12 = difference(Count, lag = 12))
c3 <- d12.lambs %>%
  autoplot(diff12) +
  labs(title = "Seasonal Lamb Differences (Lag 12)",
       y = "Year to Year Change")

c4 <- d12.lambs %>%
  ACF(diff12) %>%
  autoplot() +
  labs(title = "ACF of Seasonal Differences")
c3/c4
```



a.

Does this new data, differenced at lag12, now look like white noise? **Recall that a simple yes will not suffice. You will need to explain your conclusion.**

****Answer:** Yes, other than the first initial values which get out of the interval, the values past that seem to be small and slightly random enough to constitute white noise.

- vi. Return to the original `lambs` series and create a new column, `diff12_1`, that differences the `lambs` series at both lags 1 and 12, simultaneously (the order does not matter). Store this new data as `d12_1.lambs`.

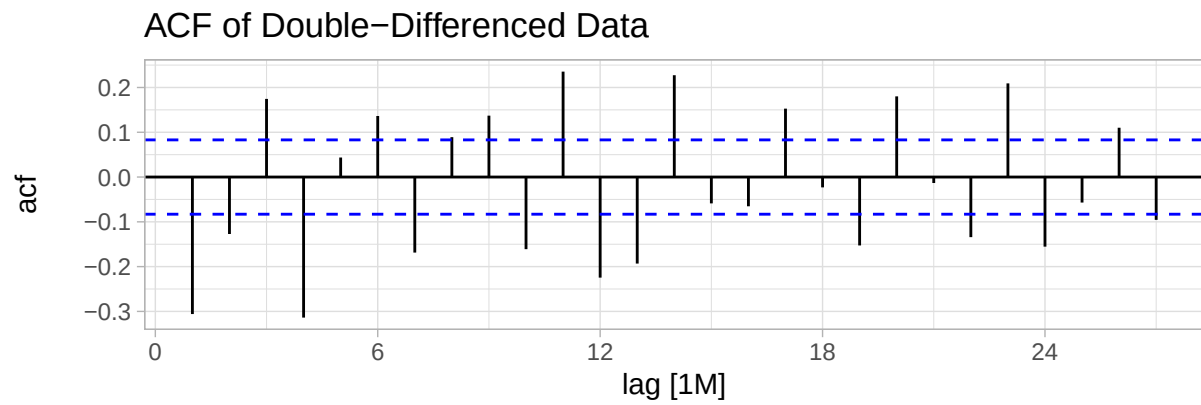
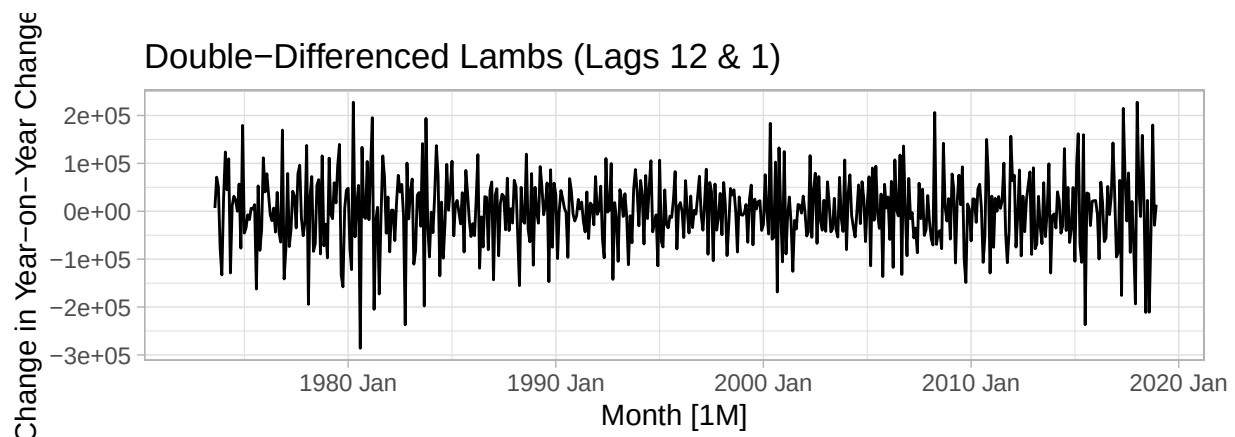
```
d12_1.lambs <- lambs %>% mutate(diff12_1 = difference(difference(Count, lag = 12), lag = 1))
```

- a. Produce an autoplot of `diff12_1` along with the associated correlogram. Does this new data, differenced at lags 1 and 12, now look like white noise? **Recall that a simple yes will not suffice. You will need to explain your conclusion.**

```
d12_1.lambs <- lambs %>%
  mutate(diff12_1 = difference(difference(Count, lag = 12), lag = 1))
c5 <- d12_1.lambs %>%
  autoplot(diff12_1) +
  labs(title = "Double-Differenced Lambs (Lags 12 & 1)",
       y = "Change in Year-on-Year Change")

c6 <- d12_1.lambs %>%
  ACF(diff12_1) %>%
  autoplot() +
  labs(title = "ACF of Double-Differenced Data")
```

```
c5/c6
```



**An-

swer: No, there is enough visually to say that this is not white noise. There is a consistent pattern of lines breaking above and below the confidence interval throughout the lags.

Question 4: Gasoline Supplied in the US

You just got your first gig as a junior forecaster at the US Energy Information Administration (EIA) on the gasoline desk. You are given the `us_gasoline` dataset, which contains weekly data on the amount of gasoline supplied in the US (measured in million barrels per day, MBd) from Week 6, 1991 to Week 3, 2017. Your task is to explore the features of this series and identify any patterns that may be present.

Your boss stopped by and apologized for giving you the weekly data instead of the monthly data, but admitted that he is unable to locate that data. You are asked to aggregate the data to monthly totals.

1. Use the `mutate()`, `index_by()`, `summarise()`, and `filter_index()` functions to create a new dataset, `us_gas_monthly`, that contains the monthly totals of gasoline supplied from **March 1991 – Dec 2016**.
Hint: Keep the Barrels column as the total amount of gasoline supplied in a month.

```
us_gas_monthly <- us_gasoline %>% mutate(Month = yearmonth(Week)) %>% index_by(Month) %>% summarise(Barrels = sum(Barrels)) %>% filter_index("1991 Mar" ~ "2016 Dec")
```

2. Next, using the following graphics functions: `autoplot()`, `gg_subseries()`, `ACF()`, explore the features of Barrels from `us_gas_online_monthly`. **Hint: You may want to use the `lag_max` of about 3 years (accounting for the data frequency) in the `ACF()` function to ensure you can see a fair bit of the pattern in the correlogram.**

```
us_gas_monthly <- us_gasoline %>%
mutate(Month = yearmonth(Week)) %>%
index_by(Month) %>%
summarise(Barrels = sum(Barrels)) %>%
filter_index("1991 Mar" ~ "2016 Dec")
z1 <- us_gas_monthly %>%
autoplot(Barrels) +
labs(title = "Monthly US Gasoline Supply",
      subtitle = "March 1991 - Dec 2016",
      y = "Total Barrels Supplied (Millions)")

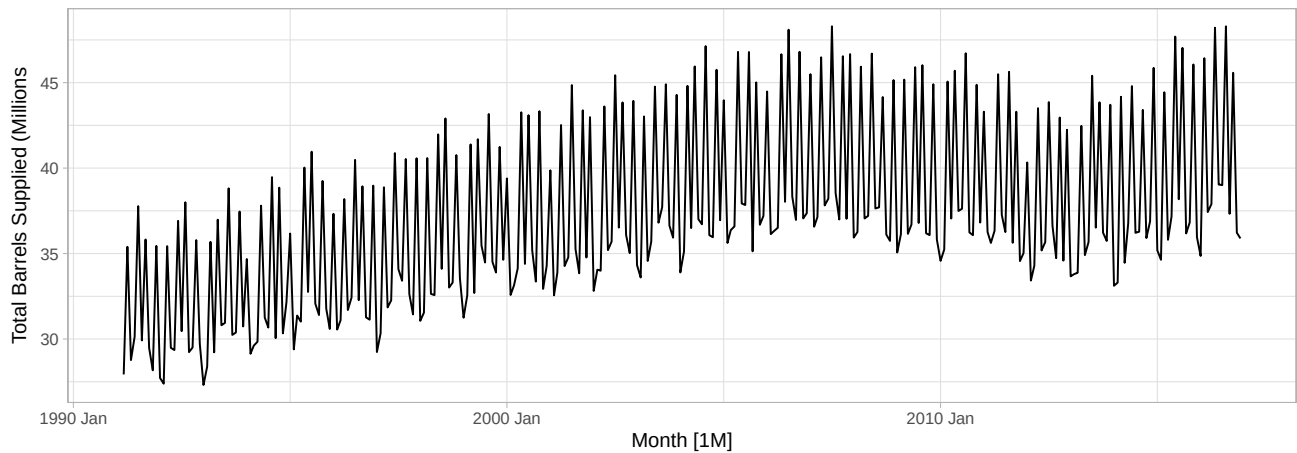
z2 <- us_gas_monthly %>%
  gg_subseries(Barrels) +
  labs(title = "Seasonal Subseries Plot",
        y = "Barrels") +
  theme(strip.text.y = element_blank())

z3 <- us_gas_monthly %>%
  ACF(Barrels, lag_max = 36) %>%
  autoplot() +
  labs(title = "ACF Plot (3-Year Lag Max)")

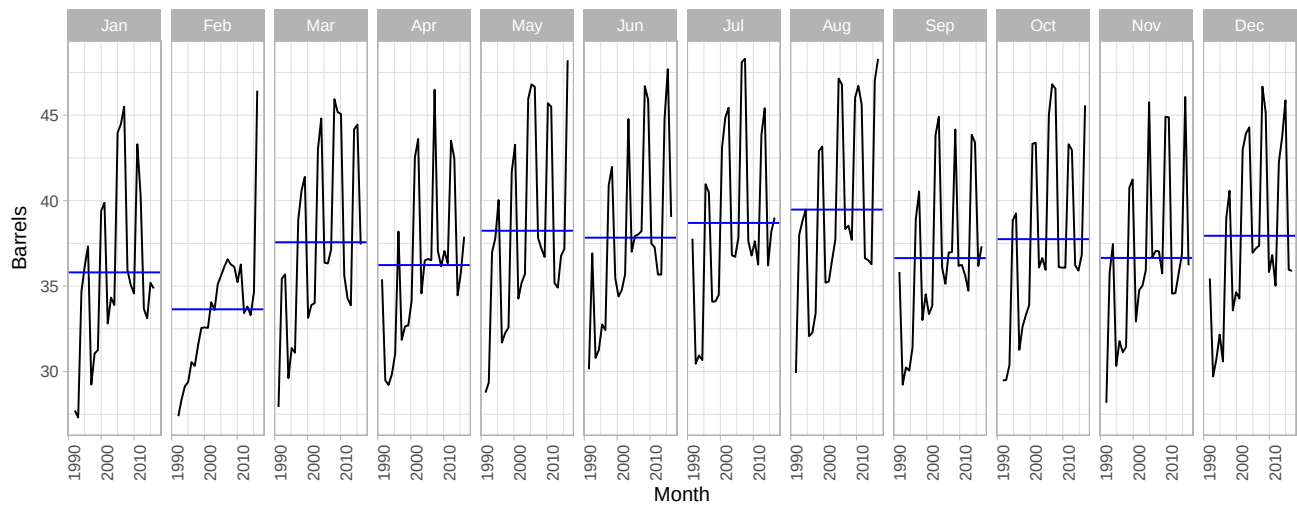
(z1/z2) / z3
```

Monthly US Gasoline Supply

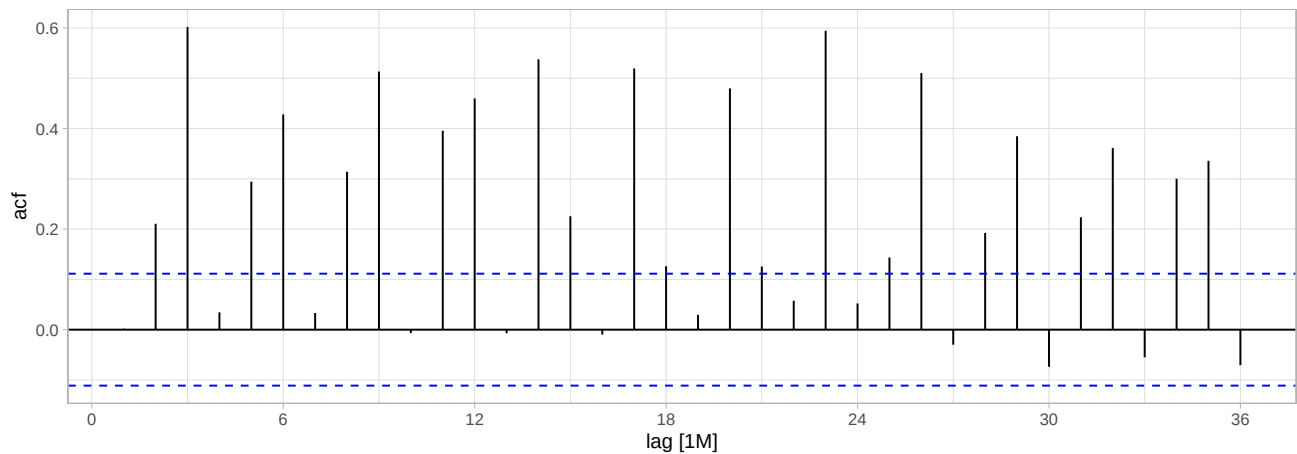
March 1991 – Dec 2016



Seasonal Subseries Plot



ACF Plot (3-Year Lag Max)



- Can you spot any seasonality, cycles, and trend?

****Answer:** The autoplot shows a trend/drift that goes upward over time that may just be attributed to higher gasoline consumption and or factors like an increased amount of cars driving.

- What do you learn about the series?

****Answer:**That this series seems to not be a white noise process as there seems to be many values outside of the interval throughout the series.

- What can you say about the seasonal patterns?

****Answer:** There is a higher supplied rate in the summer months compared to the spring, fall, and a couple of winter months. Specifically visualized in the subseries plot it can show of pattern of higher supply in summer months.